

Unit 4, Lesson 2: Points on the Coordinate Plane – Part 1



Benchmark

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x - or y -axis as the line of reflection when two ordered pairs have an opposite x - and y - coordinate.



Learning Targets

- ☐ I can plot integers using different scales.
- ☐ I can plot rational numbers in the form of decimal and fractional ordered pairs.
- ☐ I can select an appropriate scale given a set of ordered pairs.



4.2.1 Warm-Up: Interior Designer

After getting a degree in graphic design, Soledad Alzaga built her career and company as an interior designer. Interior designers make interior spaces functional, safe, and beautiful by determining space requirements and selecting decorative items such as colors, lighting, and materials. Beyond being creative and having good taste, Soledad says to own a successful interior design business, it is necessary to have the ability to work and communicate with others, be organized, have business knowledge, and to have math and measurement skills.

1. What is something that interests you that you would like to spend your career doing?

2. What skills do you think are needed to be successful in that career?



4.2.2 Refresher: Key Terms With the Coordinate Plane

Several key terms related to the coordinate plane were used throughout Lesson 1. Below, you can find definitions for each of these key terms.



The **coordinate plane** is a plane determined by two perpendicular number lines, called axes. The axes intersect at the origin. Each point in the coordinate plane is represented by a pair of coordinates that represent the direction and distance from each axis. The origin is represented by the coordinate pair, $(0, 0)$.



The **axes** of a graph are the horizontal and vertical number lines used in a coordinate plane system.



A **coordinate** is a number used to locate a point on a number line. A coordinate is one of the numbers in an ordered pair that locates a point on a coordinate plane.



The **origin** is the point of intersection of the x - and y -axes in a rectangular coordinate system, where the x -coordinate and y -coordinate are both 0.



A **quadrant** is any of the four regions separated by the axes in a coordinate plane.

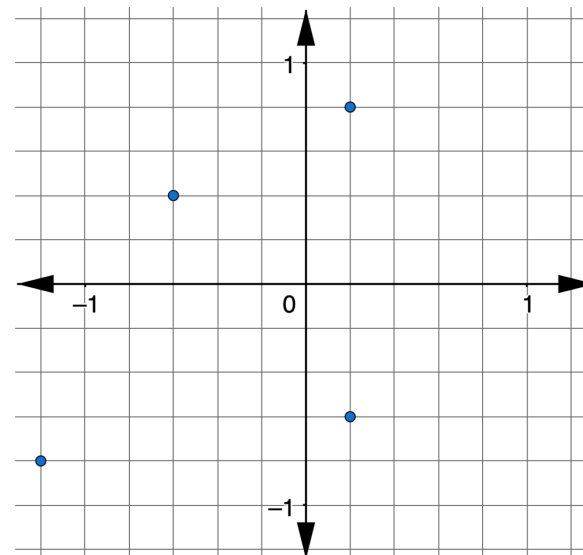


4.2.3 Guided Instruction: Plotting Points With a Fractional Scale

- Four points are shown.

Point L	Point H	Point S	Point R
$\left(\frac{1}{5}, \frac{4}{5}\right)$	$\left(-\frac{3}{5}, \frac{2}{5}\right)$	$\left(-1\frac{1}{5}, -\frac{4}{5}\right)$	$\left(\frac{1}{5}, -\frac{3}{5}\right)$

- Based on the denominators of each fraction, what scale should be used for the grid?
- Find and label the points with the given letters on the coordinate plane.



2. Another set of four points is shown.

Point M	Point L	Point C	Point A
$\left(\frac{2}{3}, \frac{1}{6}\right)$	$\left(\frac{1}{2}, -\frac{1}{3}\right)$	$\left(-\frac{5}{6}, -\frac{1}{2}\right)$	$\left(-\frac{1}{6}, 1\right)$

- a. List the denominators of each fraction.
- b. Zoe says she does not want to label her axes by halves because she will only be able to approximate the location of the thirds. She also says she does not want to label by thirds because there is no easy way to approximate the location of the halves. Christopher says she should label her axes by sixths.

Explain the benefit of using Christopher's plan.

- c. Draw and label coordinate axes with the scale you chose and plot the points.

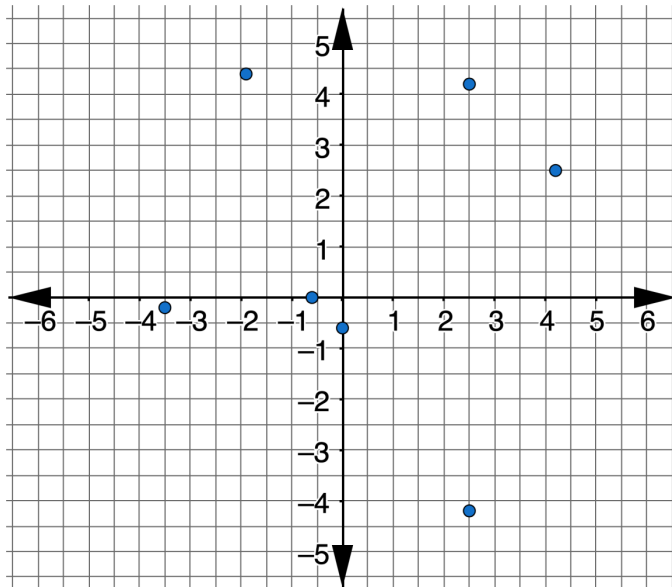


When deciding on a scale for plotting ordered pairs with rational values, consider the least common multiple (LCM) of each of the denominators to determine the proper increments.



4.2.4 Collaboration: Plotting Points With a Decimal Scale

1. A coordinate plane is shown.

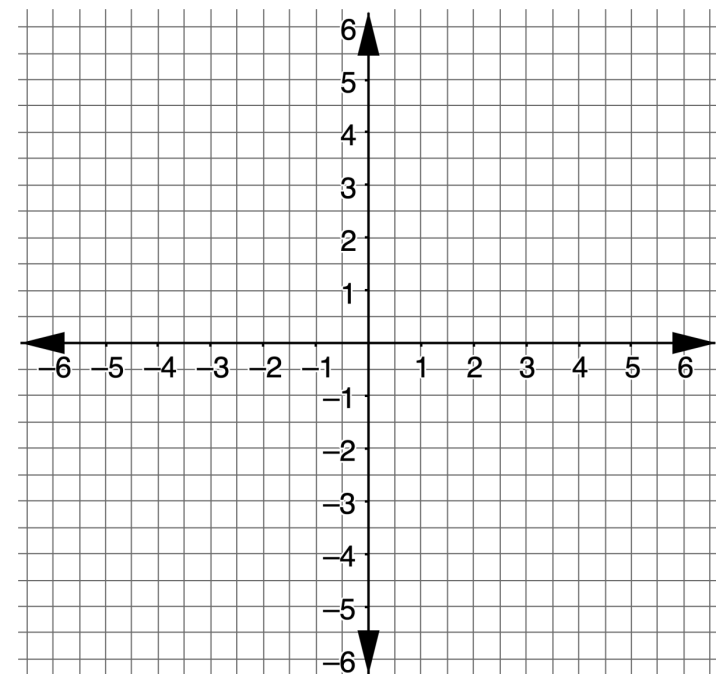


- a. Working with your partner, describe the scale that is being used for each axis.
- b. Locate each point on the coordinate plane and label it with the appropriate letter.

Point <i>G</i> (4.2, 2.5)	Point <i>B</i> (-1.9, 4.4)	Point <i>F</i> (0, -0.6)	Point <i>W</i> (-3.5, -0.2)
Point <i>E</i> (-0.6, 0)	Point <i>A</i> (2.5, 4.2)	Point <i>M</i> (2.5, -4.2)	

2. Working with your partner, plot and label the points using the given letters on the grid.

Point <i>L</i> (1.5, 2)	Point <i>B</i> (3.9, -4.5)	Point <i>G</i> (-5.5, -2.1)	Point <i>D</i> (0, 0.7)
Point <i>K</i> (-2.8, 5.7)	Point <i>S</i> (-0.8, 0)	Point <i>N</i> (-1, -3.4)	Point <i>H</i> (-3.9, 4.5)



- a. Which points were easy to plot?
- b. Which points were more difficult to plot?

3. On your own, answer the following question:

When plotting ordered pairs with decimal values, how is it best to determine the scale to use when setting up the axes on a grid?

My Reasoning

Share your reasoning with your partner and ask for feedback on any parts that were unclear or missing. Make any necessary edits to your reasoning.



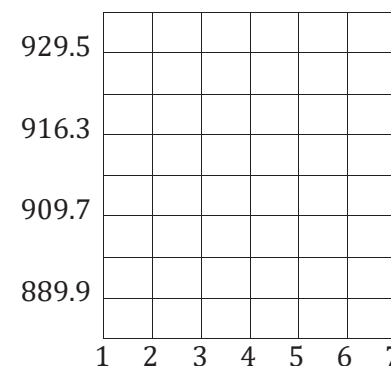
4.2.5 Collaboration: Choosing the Right Scale

When representing data, the scale of the grid is important to effectively and accurately communicate information.

1. The table shows the total Florida tourism industry employees, in thousands, for years since 2002.

Years Since 2002	Total Tourism Industry Employees (in thousands)
1	889.9
4	909.7
5	916.3
7	929.5

To plot the data, Andrea drew the coordinate grid shown.

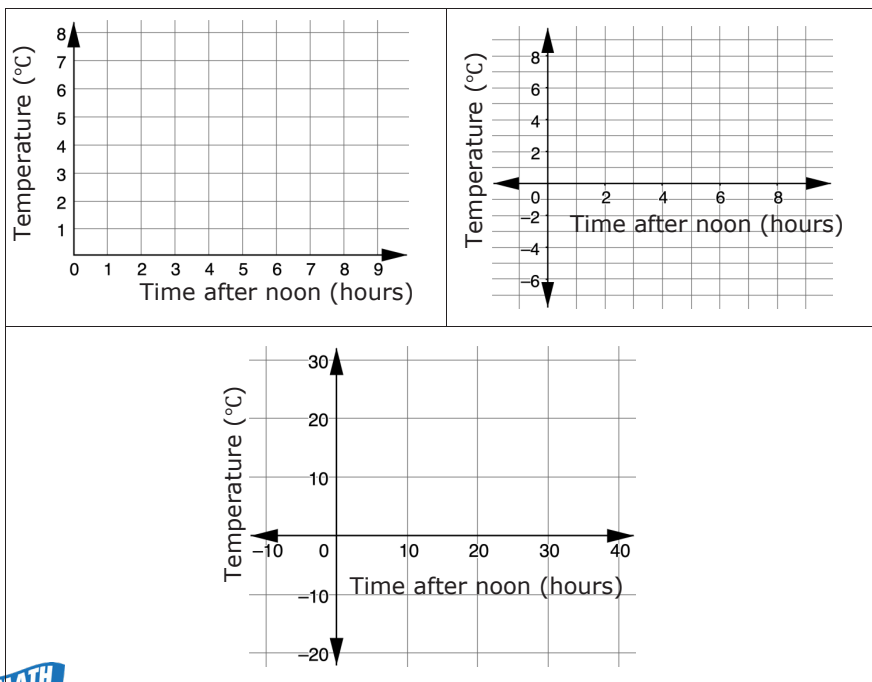


Discuss with your partner why Andrea's coordinate grid would not effectively nor accurately communicate the information about Florida tourism industry employees.

2. The data shown in the table are average autumn temperatures, in degrees Celsius ($^{\circ}\text{C}$), in Carmathen, Wales.

Time After Noon (hours)	Temperature ($^{\circ}\text{C}$)
0	5
1	4
2	3
3	2
4	1
5	-2
6	-3
7	-4
8	-4

Three coordinate grids are shown, each using different scales.



- a. Explain which coordinate grid most accurately represents the data.
- b. Explain why the other two coordinate grids may not be as accurate.



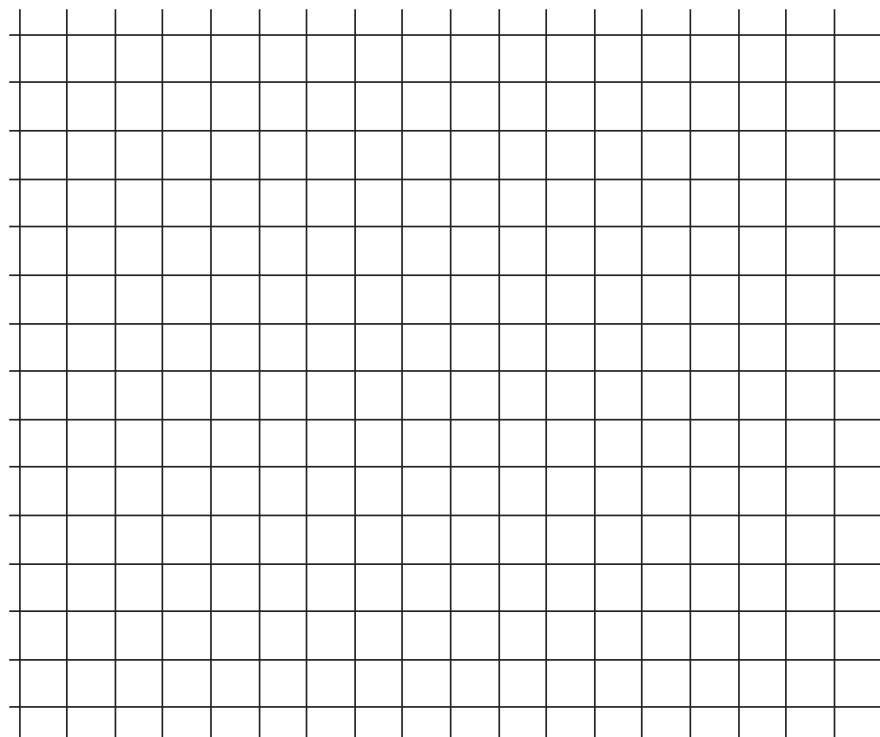
4.2.6 Try It: Plotting Points With Fractional and Decimal Scales

1. For the points $(1.75, -2.5)$ and $(-2.25, 1.5)$, draw and label coordinate axes with an appropriate scale, and plot the points.



2. Draw and label coordinate axes with an appropriate scale. Then, plot and label the points.

Point W	Point D	Point P	Point T
$\left(\frac{2}{3}, \frac{1}{4}\right)$	$\left(\frac{1}{2}, -\frac{1}{3}\right)$	$\left(-\frac{5}{3}, -\frac{1}{2}\right)$	$\left(-\frac{1}{6}, \frac{7}{4}\right)$





4.2.7 Wrap-Up: Reflection

1. Shade the arrow up to the statement that best describes your current level of understanding.



I'm so confident, I could explain plotting ordered pairs made up of fractions and/or decimals to someone else!

I can get to the right answer, but I don't understand it well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked the boxes I think I do not understand.

I tried hard and I listened, but I am finding this challenging. I checked the boxes I think I do not understand.

2. Check the skills that you currently understand.

- ☐ Determining the correct scale
- ☐ Identifying the LCM of the denominators for coordinates that include fractions
- ☐ Plotting ordered pairs that contain fractions
- ☐ Plotting ordered pairs that contain decimals

Unit 4, Lesson 3: Points on the Coordinate Plane – Part 2



Benchmark

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x - or y -axis as the line of reflection when two ordered pairs have an opposite x - and y -coordinate.



Learning Target

- ☐ I can plot rational numbers using a scale of 1.